AI usage log

Significant prompts to the AI assistant, what it produced, and whether each output was accepted, modified or rejected, with the reason and how it was verified.

28 entries	**9** accepted	**17** modified	**2** rejected	Tool: **Claude (Claude Code)**, used for research, design, code and drafting

How this log was made. It is reconstructed from the project log (decisions, what worked, what was dropped) and the working sessions; prompts and outputs are summarised, not quoted. Every entry is about the tool or its checking. The team should review the entries before submission.

No.	Date	Tool	Purpose / prompt summary	Output summary	Decision	Reason and verification
1	23 Oct 2026	Claude (Claude Code)	Frame Topic 6 (LCCA decision tool) and research the real problem, users and methods.	Proposed an IGBC credit optimizer built on life-cycle costing; quoted figures on first-cost bias (green buildings 2-17% dearer, lifetime savings above 20%).	**Modify**	Kept the IGBC centrepiece but dropped the narrow 'solar vs grid' framing. The quoted statistics come from secondary sources and are flagged for the Data Lead to verify; they are labelled as such in the guide.
2	23 Oct 2026	Claude (Claude Code)	Choose the build platform for a tool that needs charts, AI features and Monte Carlo.	Recommended Python with Streamlit and Plotly over a single HTML page or an Excel-only tool.	**Accept**	The team can read Python, and AI features and simulation are easy to add. Verified by running the app locally.
3	23 Oct 2026	Claude (Claude Code)	Write the cash flow engine: nominal rupees, one 25-year study period, re-purchases and residual value.	A vectorised engine used for single measures, sweeps and Monte Carlo runs.	**Accept**	Checked against hand-calculated unit tests (annuity, replacement, residual, escalation, degradation, solar self-use and export). Same code path for every screen.
4	23 Oct 2026	Claude (Claude Code)	Write the optimizer that picks the best combination of measures for a target IGBC level.	Exact enumeration of every legal combination (1,152 in the demo), with roof and mandatory-credit rules.	**Accept**	Provably optimal for this library size and easy to explain. Verified: test compares the optimizer's best with the full table of combinations.
5	23 Oct 2026	Claude (Claude Code)	Generate demo data (prices, savings, tariffs) for the sample office.	First draft had energy savings far too high: every measure paid back and 'buy everything' won.	**Reject**	Failed a sanity check (nothing ever flipped between routes). Data re-tuned so the trade-off is real; later rebuilt from a single sizing basis (see 15).
6	23 Oct 2026	Claude (Claude Code)	Add AI reading of vendor quotes and AI polishing of the memo, with a fallback when there is no API key.	A Claude-based extractor and memo writer, each with an offline fallback (regex parser, fixed template).	**Modify**	Offline parser read all three sample quotes correctly. The Claude path could not be tested without a key, so it was marked untested. The quote import was later removed (see 21).
7	23 Oct 2026	Claude (Claude Code)	Write the PDF decision report with reportlab.	Working report code, but it used a drawing shape (RoundRect) that does not exist in reportlab.	**Modify**	Replaced by a rectangle with rounded corners. Pages were rendered to images and inspected, which found five layout defects that tests had missed (overlapping chart labels, an orphaned heading, stray tags in cells).

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8	23 Oct 2026	Claude (Claude Code)	Choose and install supporting libraries and write table code.	Used pandas styling (needs an extra package), a version of pyarrow that clashed with NumPy, and pandas/Streamlit arguments from older versions.	**Modify**	Each caught by running the code; replaced with formatted strings, pinned versions and current argument names. Automatic page tests caught three real bugs before any demo.
9	23 Oct 2026	Claude (Claude Code)	Write the user guide with live numbers.	Guide text that hard-coded '60 tests'.	**Modify**	The count went stale at once; the guide script now counts tests by running them.
10	23 Oct 2026	Claude (Claude Code)	Simplify the interface: seven pages felt clumsy and the answer sat on page four.	Home plus four steps, a one-sentence answer on every page, three downloads instead of eleven, target level in the sidebar.	**Accept**	Reviewed by the team. Automatic tests of every screen, with and without a project, still passed.
11	23 Oct 2026	Claude (Claude Code)	Publish the project to GitHub.	Created the repository and pushed the files, including documents that contained teammate names and roll numbers.	**Modify**	Names removed (roles used instead) and the first commit replaced before the repository was made public. Open item: purge the old commit from GitHub.
12	29 Oct 2026	Claude (Claude Code)	Audit every input and output: what is unused, unverified or duplicated?	A cut list: built-up area, FAR bonus, subsidy, per-measure escalation, Pareto and MACC charts, simple payback.	**Accept**	Each item was confirmed unused by the maths or unverifiable. All tests still passed after the cuts.
13	29 Oct 2026	Claude (Claude Code)	Align the tool with the CE 725 notes (MARR, EUAC, BCR, IRR, MIRR, incremental analysis).	Mapped each on-screen metric to a chapter and proposed incremental IRR vs MARR as the headline.	**Modify**	Team decision added: rank on present worth and EUAC only; use IRR/MIRR to accept or reject (the ranking paradox, Ch 8). Course examples turned into tests (see 14).
14	29 Oct 2026	Claude (Claude Code)	Turn the course worked examples into tests of the finance functions.	Fourteen tests covering interest factors, present worth, EUAW, IRR, multiple IRR and MIRR.	**Accept**	Tests pass against the formulas. They also revealed small rounding slips in two examples in the course notes (Ch 5 and Ch 8), which are recorded in the validation report.
15	29 Oct 2026	Claude (Claude Code)	Rebuild the demo data from one consistent sizing basis, with sources.	A 12,000 m2 Pune office with typical sizes and market prices. The earlier set had a 120 TR VRF in a 12,000 m2 office, which is too small.	**Modify**	Mis-sizing corrected. Prices are still estimates; verification against recent tenders and the current tariff order is an open item.
16	29 Oct 2026	Claude (Claude Code)	Replace several input files with one Excel input workbook.	A five-sheet workbook (README, Project, Measures, IGBC_credits, IGBC_levels) with units, meanings and sources.	**Accept**	Round-trip tests: save, reopen and compare. Easier for a user to fill in than five files.
17	29 Oct 2026	Claude (Claude Code)	Package the tool as a Windows .exe for one-click start.	A PyInstaller build whose first version put app.py in the wrong folder.	**Modify**	Build script fixed, but the exe was never clicked through in a browser. Replaced by a localhost package (see 24).
18	4 Nov 2026	Claude (Claude Code)	Critique the current version and propose changes for the new demo flow.	A table of nine findings and changes. Two were partly wrong: it said the Report page had no memo and implied no step tracker existed; both already existed.	**Modify**	Findings re-checked against the code and screens; only valid ones were done. Lesson: check a critique against the real screen before acting.

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19	4 Nov 2026	Claude (Claude Code)	Add hover popups explaining every technical term (what it is, why we use it, how it is used).	A library of 25 terms with popups. In a real browser the first version was clipped under a card, and the highlight made white text in the banner unreadable.	**Modify**	Fixed stacking and banner colours; verified with an automated Chrome check: popups open by hover and keyboard, and stay on screen.
20	4 Nov 2026	Claude (Claude Code)	Remove the AI vendor-quote import (team decision).	Removed the code, sample quotes and two tests; AI left with one optional role, polishing the memo wording.	**Accept**	Keeps every number formula-driven and checkable. Tests re-run: 81 passing.
21	4 Nov 2026	Claude (Claude Code)	Check every page in a real browser.	Screenshots of every page, which exposed a layout problem in the first key-figure card (a style name used twice).	**Modify**	Style clash fixed. An early run also hit an older copy of the app on the same port, so results were re-checked on the right port.
22	4 Nov 2026	Claude (Claude Code)	Make the launcher write the live port into the browser shortcut.	Code with a line break inside a string, which is a syntax error, so the tool would not start.	**Modify**	Found by a fresh-install test of the zip, not by earlier tests. Fixed, and a test that compiles launcher.py and app.py was added.
23	4 Nov 2026	Claude (Claude Code)	Rebuild the Submission folder from the working copy.	A script that began deleting a virtual environment that was in use, leaving it damaged (a package missing).	**Modify**	Environment recreated from scratch and verified. Rule adopted: overlay files and never delete a folder that may be running.
24	4 Nov 2026	Claude (Claude Code)	Package for reviewers who run it on their own PC.	Pinned requirements; the optional AI package broke installs on deep Windows folders, so it moved to a separate file.	**Modify**	Verified by installing from the zip into a fresh environment: 82 tests pass and the app answers.
25	4 Nov 2026	Claude (Claude Code)	Use the ui-ux-pro-max skill to improve the interface.	Suggested a dark 'OLED' palette for the whole tool.	**Reject**	Not suited to a light, trust-first finance tool. Its accessibility rules were used instead: keyboard and tap access, visible focus, text for status as well as colour, reduced motion.
26	5 Nov 2026	Claude (Claude Code)	Write the three guide documents.	Included a worked LED example that said 0.5% degradation; the data says 1%.	**Modify**	Found while checking the example against the inputs; corrected and rebuilt.
27	5 Nov 2026	Claude (Claude Code)	Build the validation benchmark workbook for two alternatives.	A workbook of formulas with two mistakes: a divide-by-zero where a measure has no part, and the life-cycle cost discounted twice.	**Modify**	Found by evaluating the formulas independently and comparing with the tool; both corrected.
28	5 Nov 2026	Claude (Claude Code)	Compare the benchmark with the tool for VRF vs chiller.	A comparison of ten metrics for each alternative plus three incremental figures.	**Accept**	All comparisons agree to better than 0.0000001 rupees; a second closed-form method agrees too. Recorded in the validation report.